

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Technical Presentation

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	<i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.</i>		
Student Name:	S.Roshni	Reg. No.:	192525185
Section / Batch:	AI&ML	Date:	15/07/2026

Code of Conduct

*I S.Roshni(192525185) _____ (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this
assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: S.Roshni_____

Instructions

- This assessment contains technical presentation. Total: 100 Marks.
- When a student delivers a technical presentation on a data structures topic, evaluation should be based on the following requirements: Concept explanation, Real-world application and Clarity of presentation.
- Demonstrate clear understanding, not just reading slides
- Use of tools: PowerPoint / Google Slides.
- Duration: 7–10 minutes presentation + 3 minutes Q&A.

Section / Topic		Focus	Q Nos	Marks
Data Structure		Real-Time Applications	1	100
GRAND TOTAL:			100 Marks	

Annexure A – Technical Presentation

A web browser supports navigating backward through previously visited pages. Recommend a suitable ADT and prepare a technical presentation explaining how it improves performance efficiency. (100 Marks)

(OR)

A music streaming application maintains a playlist where songs can be added, removed, or reordered dynamically. Recommend a suitable data structure and justify based on flexibility and efficiency. Prepare a technical Presentation. (100 Marks)

(OR)

A metro rail network maps stations and routes to determine reachable stations from a source station. Suggest a suitable data structure for representation and justify your answer. (100 Mark)

(OR)

A calculator application evaluates arithmetic expressions entered by users. Recommend a suitable data structure and justify based on expression processing efficiency. (100 Marks)

(OR)

A cafeteria token system serves customers strictly in the order of arrival. Identify the suitable ADT and justify why it ensures fairness. (100 Marks)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	20	Demonstrates deep and accurate understanding of data structure with insights and connections	Shows clear understanding with minor gaps	Basic understanding; some misconceptions present	Limited or incorrect understanding
Content Organization	30	Logical, well-structured, smooth flow; strong introduction and conclusion	Mostly organized with clear flow	Some organization but lacks clarity or transitions	Disorganized, difficult to follow
Technical Depth	20	Includes algorithms, complexity analysis, and advanced insights	Includes algorithm and basic complexity analysis	Limited technical details; missing depth	Minimal or no technical content
PowerPoint Slides	20	Excellent design, clear visuals, enhances understanding	Good slides with minor issues	Basic slides; somewhat cluttered or unclear	Poor slides or no visual support
Communication Skills	10	Clear, confident, engaging delivery; excellent articulation	Clear delivery with minor hesitation	Some hesitation; unclear in parts	Poor communication; difficult to understand



Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



WEB BROWSER BACKWARD NAVIGATION

Using Stack (Abstract Data Type)



Data Structures — Course Presentation

Student Name: S. Roshni

Register Number: 192525185

Course: Data Structures

TOP (Back removes this first)

Gmail

YouTube

Google

Home



Abstract

A web browser uses the Stack Abstract Data Type (ADT) to manage backward navigation efficiently. Each visited webpage is pushed onto the stack, and pressing the Back button pops the most recently visited page, following the Last In, First Out (LIFO) principle. This approach provides constant-time navigation ($O(1)$), improves browser responsiveness, reduces processing overhead, and offers a simple yet efficient solution widely used in modern web browsers.



What is a Stack?

A linear data structure that follows the LIFO principle — Last In, First Out.



Push (Insert)

Adds a new page to the top of the stack when the user visits it.



Pop (Remove)

Removes the top page when the user clicks the Back button.



Peek (View Top)

Looks at the current top page without removing it.

Stack Example

TOP
→

Gmail

YouTube

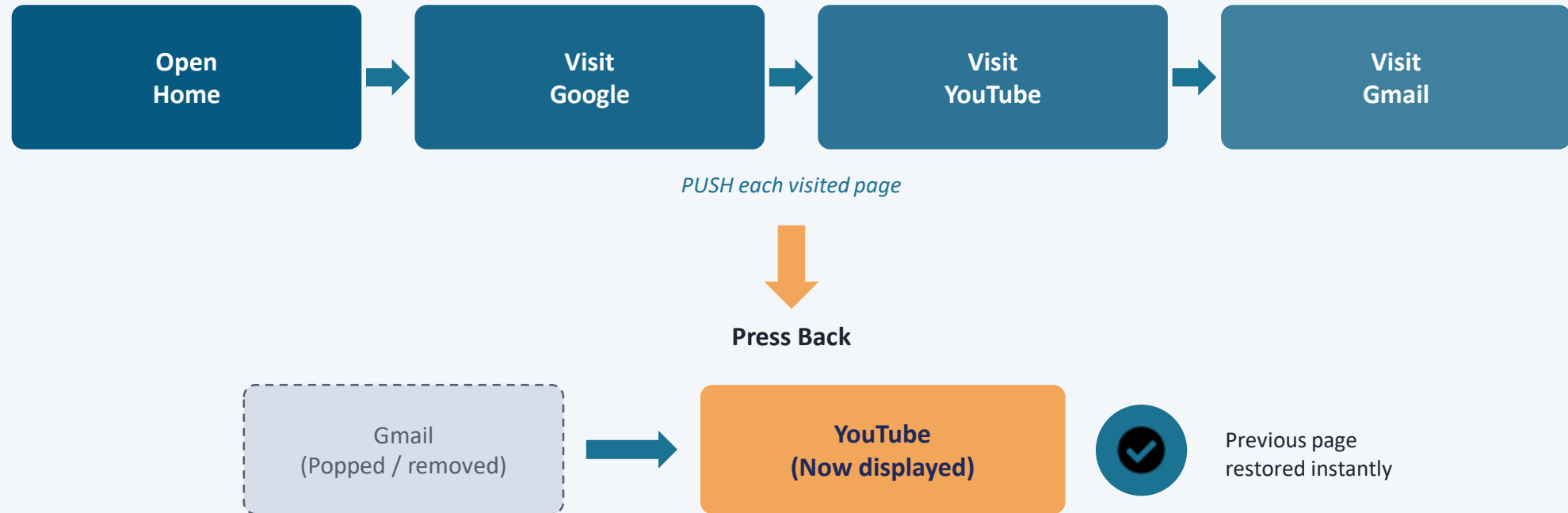
Google

Home

When Back is pressed, Gmail is removed first.

Working of Browser Back Navigation

Every new page is Pushed. When Back is clicked, the Top page is Popped.



Real-World Application

Where is Stack Used?



Google Chrome



Microsoft Edge



Mozilla Firefox



Safari



Mobile Browsers

Other Uses of Stack



Undo Operation



PDF Reader



Image Viewer



File Explorer

Almost every modern browser and OS-level tool relies on the Stack ADT for fast, ordered undo/back behavior.

Performance Efficiency

Why Stack?

Operation	Time Complexity
Push	$O(1)$
Pop	$O(1)$
Peek	$O(1)$
Memory Complexity	$O(n)$

Advantages

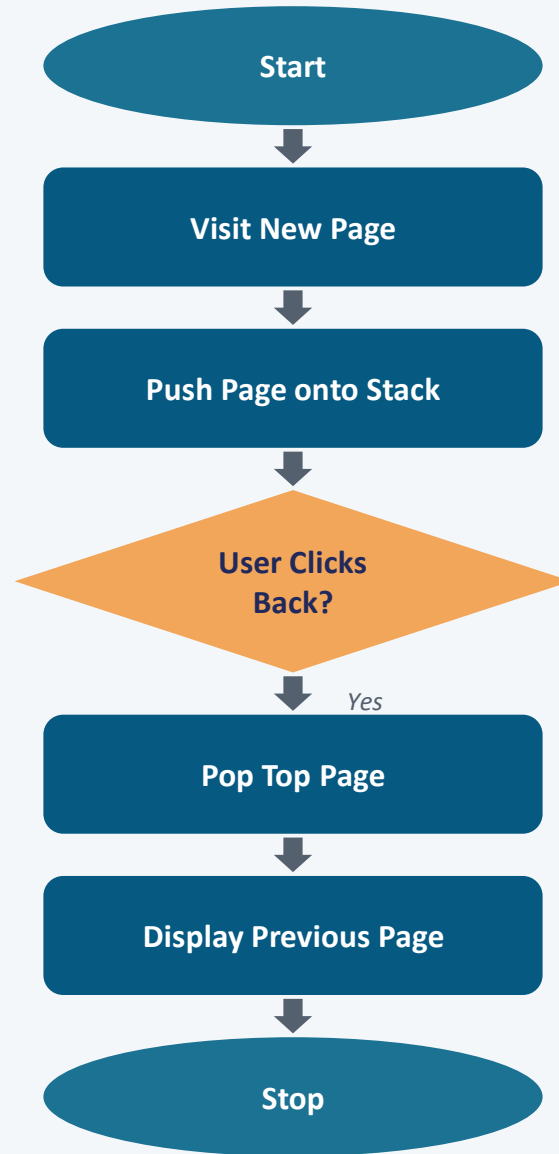
- ✓ Fast constant-time operations
- ✓ Simple to implement
- ✓ Less memory overhead
- ✓ Improves browser performance



Constant-Time Operations

Since Push and Pop operations take constant time, browser navigation remains fast even when many webpages have been visited.

Algorithm



How it works



Each new webpage the user visits is pushed onto the stack.



The stack keeps pages in visit order, with the latest on top.



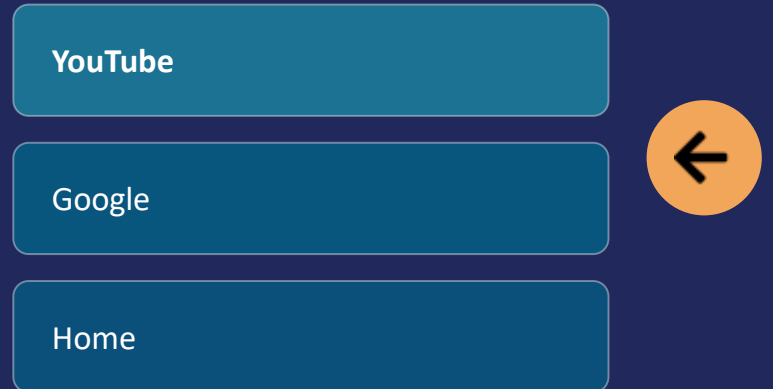
When the user clicks Back, the top page is popped off.



The next page below becomes the new top and is displayed.

Conclusion

- ✓ Stack follows the LIFO (Last In, First Out) principle.
- ✓ It is the best ADT for browser backward navigation.
- ✓ Push and Pop operations take $O(1)$ constant time.
- ✓ It provides fast, efficient, and user-friendly navigation.
- ✓ Stack is widely used in real-world browsers and applications.



Thank You!

To conclude, the Stack ADT is the ideal data structure for browser backward navigation. Its LIFO behavior and constant-time operations make browsing efficient, reliable, and responsive.

Thank You!

FOR YOUR ATTENTION



STACK ADT – POWERING
SMARTER NAVIGATION

